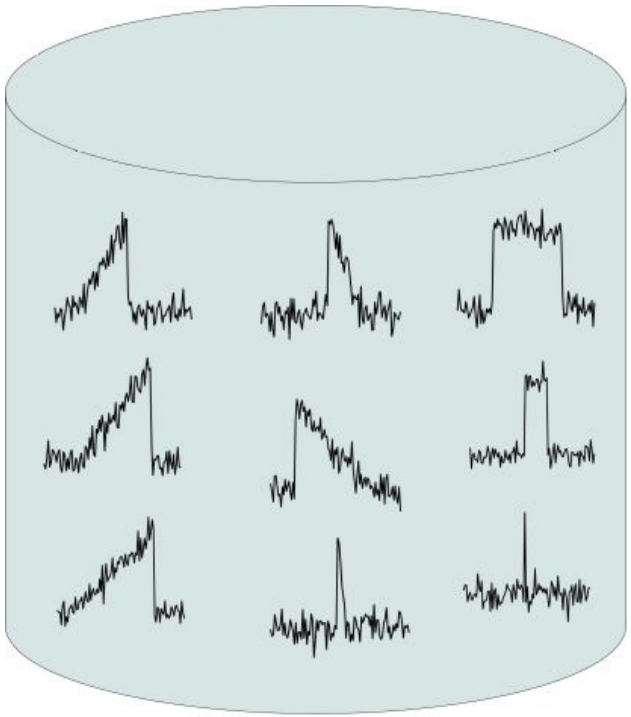
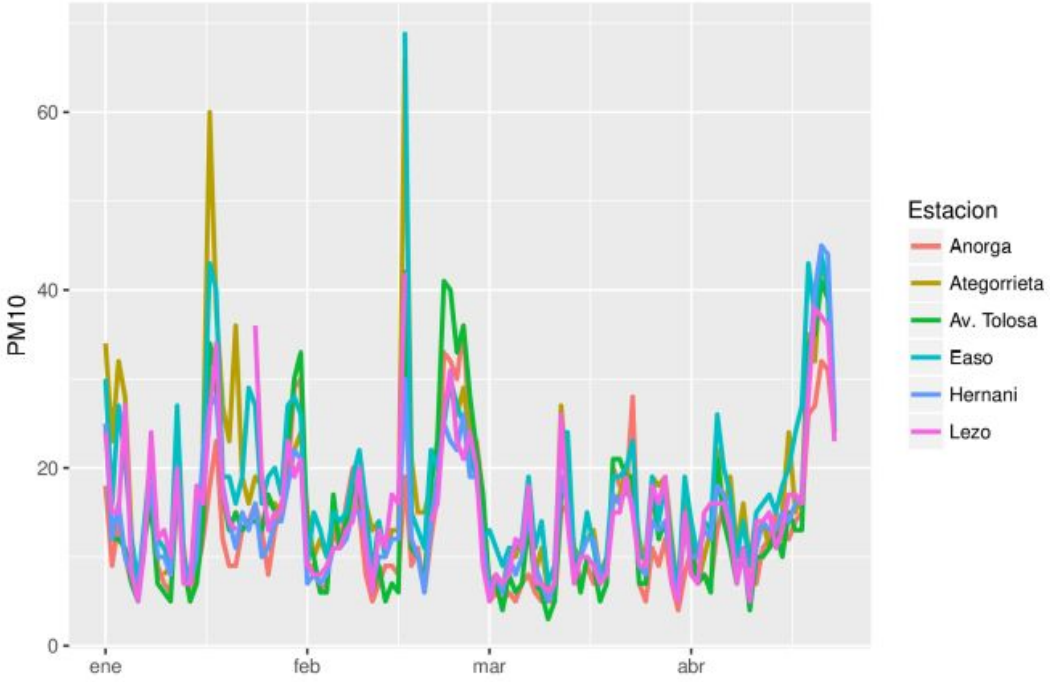
DISTANCES FOR TIME SERIES

A database of time series

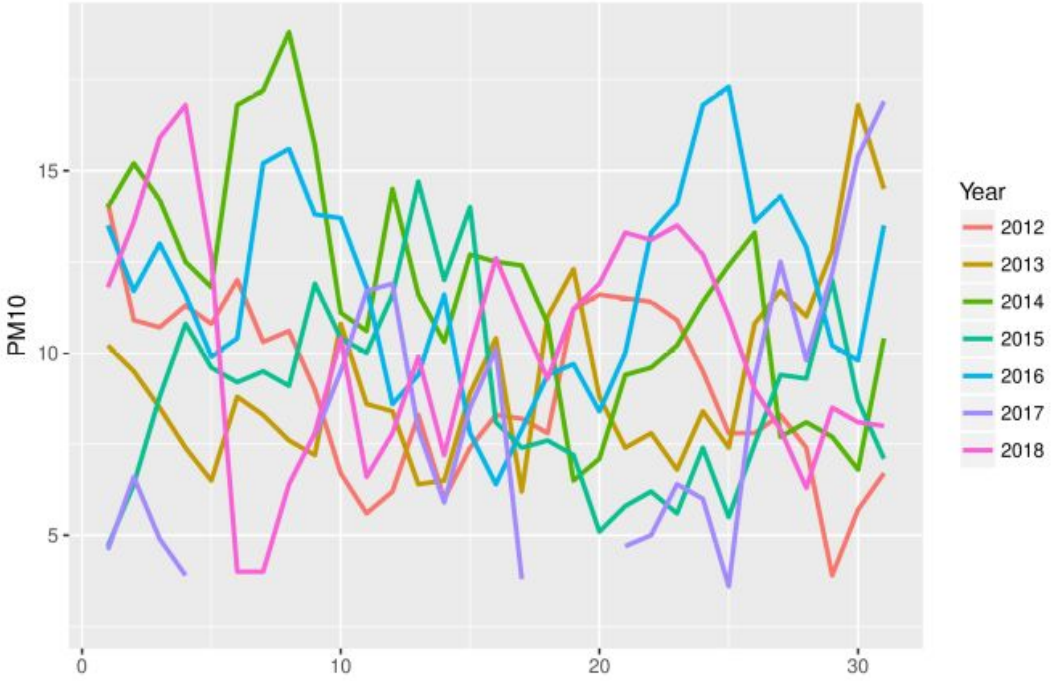
- A set of unordered time series.
- The lengths of the time series can vary.



Daily air quality (PI0) in different stations

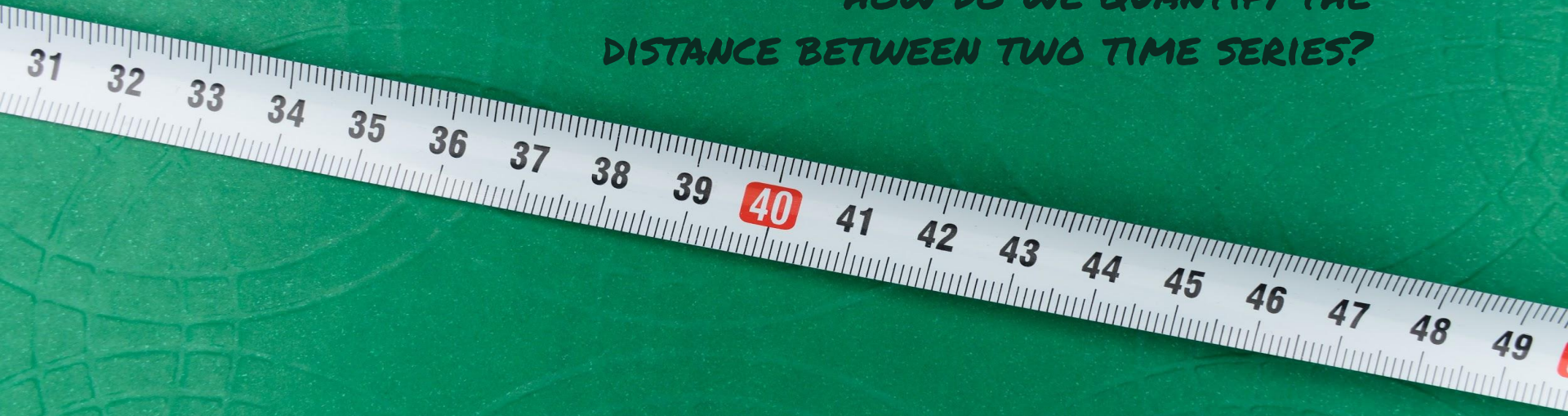


Daily temperature in Hernani in January

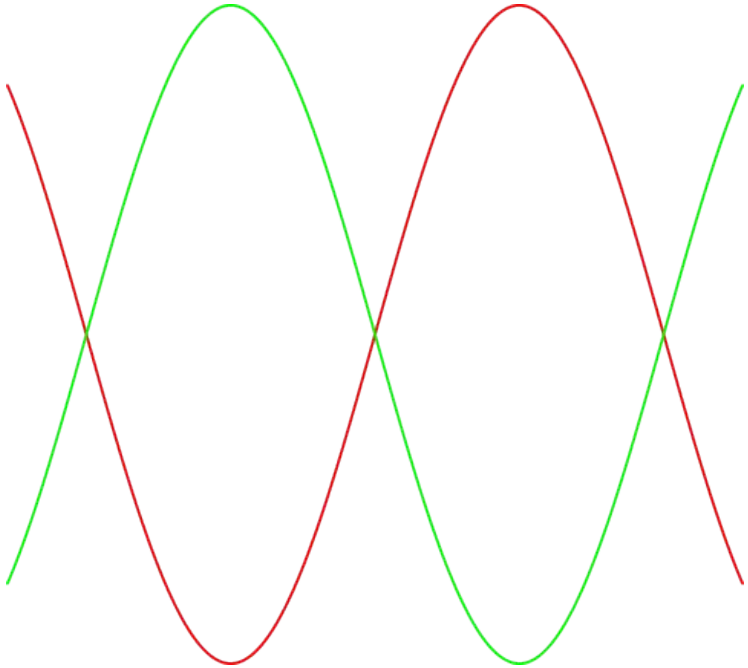


MANY ALGORITHMS ARE BASED ON DISTANCES BUT...

HOW DO WE QUANTIFY THE DISTANCE BETWEEN TWO TIME SERIES?

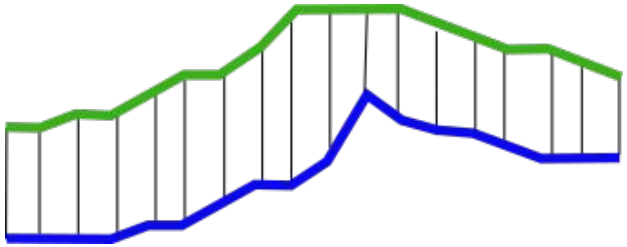


Are these two time series similar?



Distances between raw time series

Fixed distances



Flexible distances

Euclidean distance

$$D(X,Y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

✓ Very simple and easy to calculate.

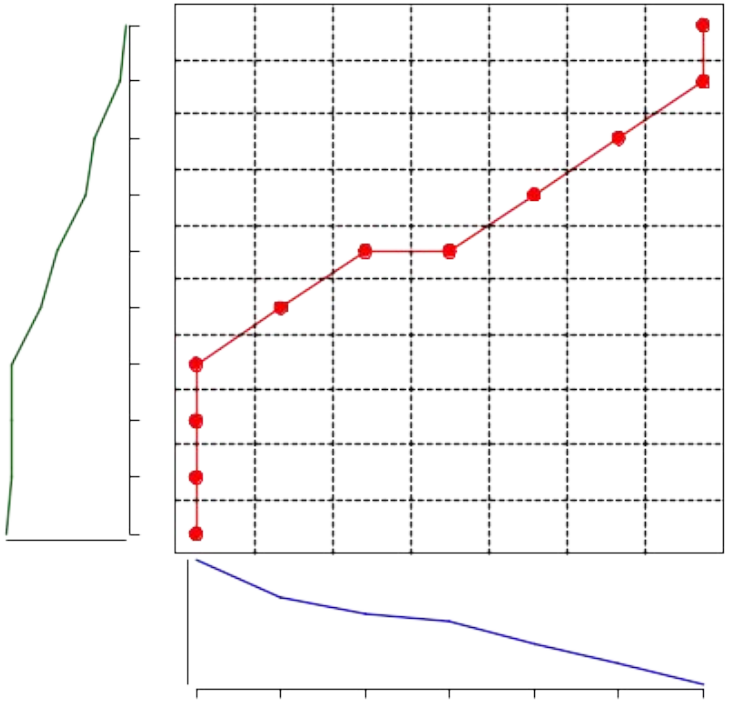
✓ Very low computational cost.

✗ Can only be used to compare series of the same length.

✗ Temporal correlation is not taken into account.

✗ Very susceptible to noise.

Dynamic Time Warping



- ✓ Takes temporality into account.
- ✓ Can be used to compare series of different lengths
- ✗ High computational cost
- ✗ Not a metric (does not fulfil triangle inequality)

Dynamic Time Warping: recursive formula

$$DTW(X,Y) = \begin{cases} 0 & \text{if } M-1 = N-1 = 0 \\ \inf & \text{if } M-1 = 0 \text{ or } N-1 = 0 \\ d(x_0, y_0) + min\{DTW(Rest(X), Rest(Y)), \\ DTW(Rest(X), Y), DTW(X, Rest(Y))\} & \text{otherwise} \end{cases}$$

Dynamic Time Warping: dynamic programming

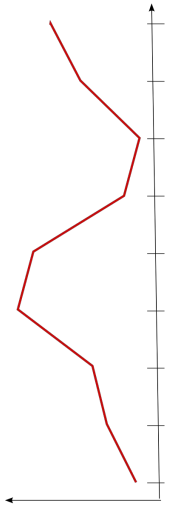
Algorithm 1 Dynamic Time Warping calculation.

Require: $X = \{x_0, ..., x_{N-1}\}, Y = \{y_0, ..., y_{M-1}\}$
Ensure: $costMatrix = matrix(N+1, M+1) * 0$
  for $i = 1, ..., N$ **do**
    $costMatrix(i, 0) = 10000$
  end for
  for $j = 1, ..., M$ **do**
    $costMatrix(0, j) = 10000$
  end for
  for $i = 1, ..., N$ **do**
    for $j = 1, ..., M$ **do**

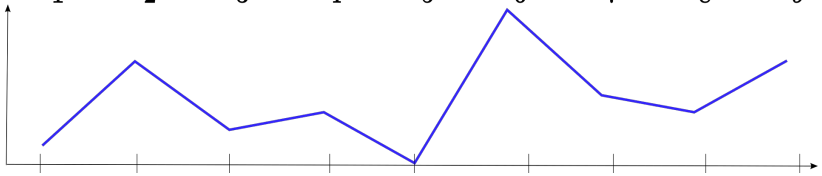
$$costMatrix(i, j) = d(x_i, y_j) + min \begin{cases} costMatrix(i-1, j) \\ costMatrix(i, j-1) \\ costMatrix(i-1, j-1) \end{cases}$$

    end for
  end for
  return $costmatrix(N, M)$

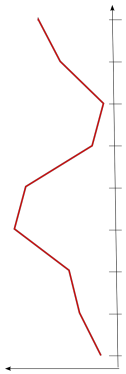
Dynamic Time Warping: example



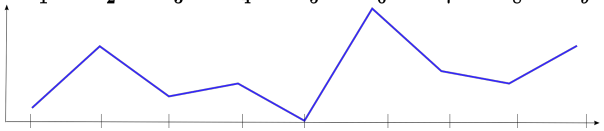
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
y_9	6	1	5	4	7	2	3	4	1
y_8	4	1	3	2	5	4	1	2	1
y_7	0	5	1	2	1	8	3	2	5
y_6	1	4	0	1	2	7	2	1	4
y_5	7	2	6	5	8	1	4	5	2
y_4	8	3	7	6	9	0	5	6	3
y_3	3	2	2	1	4	5	0	1	2
y_2	2	3	1	0	3	6	1	0	3
y_1	0	5	1	2	1	8	3	2	5

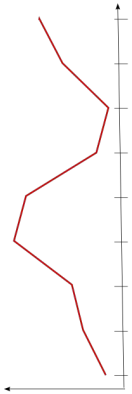


Dynamic Time Warping: example

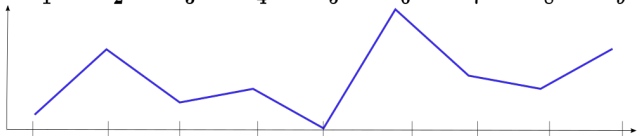


	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
y_9	6	1	5	4	7	2	3	4	1
y_8	4	1	3	2	5	4	1	2	1
y_7	0	5	1	2	1	8	3	2	5
y_6	1	4	0	1	2	7	2	1	4
y_5	7	2	6	5	8	1	4	5	2
y_4	8	3	7	6	9	0	5	6	3
y_3	3	2	2	1	4	5	0	1	2
y_2	2	3	1	0	3	6	1	0	3
y_1	0	5	1	2	1	8	3	2	5

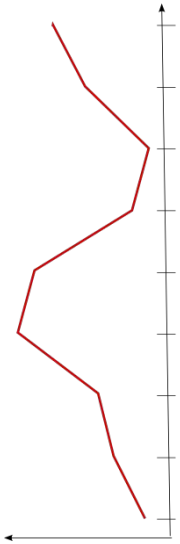




	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
y_9	31	20	18	16	19	17	17	18	15
y_8	25	19	13	23	16	15	14	15	14
y_7	21	18	10	11	11	19	14	13	17
y_6	21	13	9	10	12	16	11	12	16
y_5	20	9	13	16	19	9	12	17	18
y_4	13	7	11	11	14	8	13	18	16
y_3	5	4	5	5	8	12	12	13	15
y_2	2	3	4	4	7	13	14	14	17
y_1	0	5	6	8	9	17	20	22	25



Dynamic Time Warping: example



	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
y_9	31	20	18	16	19	17	17	18	15
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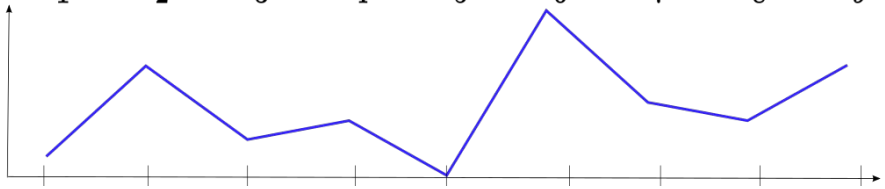
y_7	21	18	10	11	11	19	14	13	17
y_6	21	13	9	10	12	16	11	12	16
y_5	20	9	13	16	19	9	12	17	18
y_4	13	7	11	11	14	8	13	18	16
y_3	5	4	5	5	8	12	12	13	15
y_2	2	3	4	4	7	13	14	14	17
y_1	0	5	6	8	9	17	20	22	25



EDIT DISTANCES

Smallest number of INSERT, DELETE, SUBSTITUTE operators that must be applied to convert one sequence into the other.

```
I N T E * N T I O N
| | | | | | | | | |
* E X E C U T I O N
d s s   i s
```

NOTE: To work with time series (real values sequences), different adaptations have been proposed, EDR, ERP, etc.

EDIT DISTANCES

- ✓ Temporality is considered.
- ✓ Can be used to deal with series of different lengths.
- ✓ Less susceptible to noise and outliers.

- ✗ High computational cost.
- ✗ They are not always metrics.

Software

- Distances for time series:

  TSclust
  TSdist
  dtw

Software

- Distances for time series:

  sklearn.metrics
  aeon

Bibliography

- Philippe Esling and Carlos Agon. 2012. Time-series data mining. ACM Comput. Surv. 45, 1, Article 12 (December 2012), 34 pages.

- Mori, U., Mendiburu, A., & Lozano, J. A. (2016). Distance Measures for Time Series in R: The TSdist Package. R J., 8(2), 451.

- Romain Tavernard. An introduction to dynamic time warping. https://rtavenar.github.io/blog/dtw.html